

DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY, LONERE

Rescheduled Supplementary Summer Examination – 2025

Course: B.Tech (Computer Engineering / Computer Science & Engineering / Allied)

Semester : V

Subject Code & Name: Database Management System (BTCOC501)

Max Marks: 60

Date: 18/08/2025

Duration: 3 Hr.

Instructions to the Students:

- 1 Each question carries 12 marks.
- 2 Question No. 1 will be compulsory and include objective-type questions.
- 3 Candidates are required to attempt any four questions from Question No. 2 to Question No. 6.
- 4 The level of question/expected answer as per OBE or the Course Outcome (CO) on which the question is based is mentioned in () in front of the question.
- 5 Use of non-programmable scientific calculators is allowed.
- 6 Assume suitable data wherever necessary and mention it clearly.

Marks

Q. 1 Objective type questions. (Compulsory Question)

12

- 1 The primary purpose of a database system is to _____. 1
 - a) Store large text files
 - b) Provide efficient data management
 - c) Generate operating systems
 - d) Run hardware devices
- 2 Which of the following is a weak entity? 1
 - a) Student
 - b) Employee
 - c) Dependent
 - d) Course
- 3 In ER diagrams, mapping cardinality specifies: 1
 - a) Attributes of entity
 - b) Number of entities in a relationship
 - c) Storage size
 - d) File format
- 4 Match the following: 1

1. Primary Key	A. Unique identifier
2. Candidate Key	B. Minimal super key
3. Super Key	C. Set of attributes uniquely identifying tuples
4. Foreign Key	D. Referential integrity

 - a) 1-A, 2-B, 3-C, 4-D
 - b) 1-B, 2-C, 3-D, 4-A
 - c) 1-C, 2-D, 3-B, 4-A
 - d) 1-D, 2-A, 3-C, 4-B
- 5 Tuple relational calculus is _____. 1
 - a) Procedural
 - b) Non-procedural
 - c) Data definition language
 - d) Encryption technique
- 6 The SQL clause used to group tuples is: 1
 - a) ORDER BY
 - b) GROUP BY
 - c) HAVING
 - d) DISTINCT

- 7 Normalization is used to _____. 1
- a) Increase redundancy b) Minimize redundancy
c) Reduce concurrency d) Improve hardware speed
- 8 A relation in 1NF does not contain: 1
- a) Atomic values b) Multi-valued attributes
c) Candidate keys d) Primary keys
- 9 A schedule that preserves transaction results as if executed serially is called: 1
- a) Non-serializable b) Serializable c) Concurrent d) Cascading
- 10 Which of the following is NOT a concurrency control protocol? 1
- a) Lock-based b) Timestamp-based
c) Two-phase commit d) Deadlock avoidance
- 11 Match the following: 1
- | | |
|----------------|----------------------------|
| 1. Atomicity | A. All or nothing |
| 2. Consistency | B. Preserves rules |
| 3. Isolation | C. Executes independently |
| 4. Durability | D. Results survive failure |
- a) 1-A, 2-B, 3-C, 4-D b) 1-B, 2-C, 3-D, 4-A
c) 1-C, 2-D, 3-A, 4-B d) 1-D, 2-A, 3-B, 4-C
- 12 A _____ is a logical unit of database processing that includes one or more operations. 1
- a) Tuple b) Transaction c) Query d) Relation
- Q. 2 Solve the following.** 12
- A) Explain the difference between primary key, candidate key, and super key with examples. 6
- B) Define database system applications with examples. Differentiate between strong and weak entity sets in ER modeling. 6
- Q.3 Solve the following.** 12
- A) List the fundamental operations in relational algebra. 6
- Write relational algebra queries for:
- a) Retrieve names of employees earning more than 50,000.
b) Find all students enrolled in "Database Systems".
- B) Explain schema, instance, and relation with an example. Differentiate between relational algebra and relational calculus. 6
- Q. 4 Solve Any Two of the following.** 12
- A) List the different types of SQL commands with examples. Define SQL joins and explain types with syntax. 6
- B) Explain any FOUR built-in types the SQL. Explain aggregate functions in SQL with suitable 6

examples.

Write SQL queries for the table consists of Employee:

- a) Find names of employees working in "Sales" department.
 - b) List the average salary by department.
 - C) The following are SQL queries based on a typical "university schema," which commonly includes tables such as course, department, instructor, student, takes, and section. 6
 - a) Find the titles of courses in the Comp. Sci. department that have 3 credits.
 - b) Find the IDs of all students who were taught by an instructor named Einstein; make sure there are no duplicates in the result.
 - c) Find the highest salary of any instructor.
 - d) Find all instructors earning the highest salary (there may be more than one with the same salary).
- Q.5 Solve Any Two of the following.** 12
- A) State the features of a good relational design. Define functional dependency with example. 6
 - B) Why are certain functional dependencies called trivial functional dependencies? Normalize the given relation R(A, B, C, D, E) with FDs: $A \rightarrow B$, $B \rightarrow C$, $C \rightarrow D$, $D \rightarrow E$ up to BCNF. 6
 - C) Compute the closure of the following set F of functional dependencies for relation schema r (A, B, C, D, E). $A \rightarrow BC$, $CD \rightarrow E$, $B \rightarrow D$, $E \rightarrow A$ 6
List the candidate keys for R.
- Q.6 Solve Any Two of the following.** 12
- A) Explain lock-based concurrency control with examples. State the ACID properties of a transaction. 6
 - B) Show how timestamp-based protocol can avoid conflicts between two sample transactions. Differentiate between static and dynamic deadlock handling techniques. 6
 - C) Draw an ER diagram for a university system with entities: Student (Student_ID [PK], Name, DOB, Email, Phone), Course (Course_ID [PK], Course_Name, Credits, Department, Semester), Instructor (Instructor_ID [PK], Name, Qualification, Department, Email) having the relations as Enrollment, Teaches, and advises. 6

*** End ***